



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Addis Ababa University
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PLACENTA ACCRETA SPECTRUM

Dr. Hussen T (MD, OBGYN Resident)

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OUTLINE

- Introduction
- Epidemiology
- Risk factors
- Pathogenesis
- Diagnosis
- Management
- Summary

INTRODUCTION

- PAS: describes **aberrant placentation**
 - Abnormal placental villous attachment/invasion into the myometrium beyond the physiologic decidual–myometrial junction zone.
 - When PAS is present, the **failure of the entire placenta to separate** normally from the uterine wall after delivery is typically accompanied by **massive obstetric hemorrhage**.

Cont'd

- PAS: is one of the most dangerous conditions in pregnancy.
- The risk of life-threatening bleeding is present throughout pregnancy.
- Like 3rd-TM deliveries, abnormal placentation during the 2nd TM:
 - Suspected based on abnormal imaging.
 - Diagnosed at the time of termination resulting in heavy bleeding.

EPIDEMIOLOGY

- PAS is still relatively uncommon with a prevalence of ~1 per 800 births,
 - It is likely to become more common as CD rates increase around the world.

RISK FACTORS

NO PP, ONLY PRIOR CS SCAR

- 1-scar → 0.2%
- 2-scar → 0.3%
- 3-scar → 0.6%
- 4-scar → 2.1%
- 5-scar → 2.3%
- 6-scar → 6.7%

PP WITH PRIOR CS SCAR

- The most important risk factor for PAS, found in **50%** of all cases, is **PP**.
- The risk of PAS ↑ with both the **No of prior CDs** & the **presence of PP**.
- A large multicenter US cohort study found that for women presenting with PP & prior CDs, the risk of accreta is:
 - PP wz **no scar**→03%
 - PP wz 1scar →11%
 - PP wz 2scar → 40%
 - PP wz 3scar →61%
 - PP wz ≥4scar →67%

TABLE 20.2 Risk of Placenta Accreta Spectrum (PAS) With Placenta Previa and Prior Cesarean Delivery

Number of Prior Cesarean Deliveries	PAS Risk (%)
0	3
1	11
2	40
3	61
≥4	67

Other Risk Factors

- Any form of damage to the integrity of the **endometrial-myometrial junctional zone** following:
 - MVA, D & C,
 - Myomectomy,
 - Endometrial resection(Removal of adhesion)
 - Endometrial ablation
 - Postpartum Endometritis
 - ART(Embryo transfer)
- Uterine anomalies, Adenomyosis, **submucous fibroids,, AMA, MP & Multiple pregnancy** have also been associated with PAS.

PATHOGENESIS

THE 2-KEY FACTORS IN THE PATHOGENESIS OF PP ARE:

ABNORMAL VASCULARIZATION

- Suboptimal vascularization & tissue oxygenation of an **area of scar** →
 1. **1^o defect of trophoblast biology** → excessive invasion of the myometrium, &
 2. **2^o defect of decidualization** (Abnormal decidualization) →
 - Preferential attachment of the blastocyst to scar tissue &
 - Deeper invasion of the villous tissue & Trophoblast infiltration
- Uterine artery resistance ↑, and the volume of uterine blood flow ↓ in women with a previous CD compared to women with a previous vaginal birth.

DIAGNOSIS

DIAGNOSIS (1ST TM PAS)

TABLE 2

Definitions of placenta accreta spectrum markers in the first trimester of pregnancy

Marker	Definition
Cesarean scar pregnancy	Gestational sac implantation in part or totally within the cesarean scar.
	Gestational sac may have a teardrop or triangular shape.
Low implantation pregnancy	Gestational sac located close to the internal cervical os (up to 8 6/7 weeks of gestation) and/or placental implantation located posterior to a partially filled maternal bladder (up to 13 6/7 weeks of gestation).

Shainker. Special Report of the SMFM: Definition of markers and ultrasound examination in pregnancies at risk of PAS. Am J Obstet Gynecol 2021.

DIAGNOSIS (2nd/3rd TM)

① Placental bulk: **lacunae**

- Grade 2+ (4-6 lacunae)
- Turbulent lacunar blood flow (PSV>15 cm/s)

② Abnormal utero-placental interface

- Loss of clear zone*
- Myometrial thinning (<1mm)
- ↑d vascularity on color doppler

③ Abnormal Utero-vesical interface

- UV-hypervascularity
- Bridging vessels,
- Interruption of the bladder wall (**Disruption** of the BUSI)

④ Miscellaneous markers

- Placental bulge,
- Exophytic placental mass, and
- Cervical vascular extension

MRI in diagnosing PAS

- MRI may add information particularly in cases of **posterior placenta** in assessing :
 - **Topography** (areas of abnormal placentation)
 - **Depth** of myometrial invasion.
- Sensitivity-**94%**, Specificity-84%

TABLE 20.3 Placenta Accreta Spectrum (PAS) International Federation of Gynecology and Obstetrics Grading System

PAS Grade	Histopathologic Findings
Grade 1	Decidua absent between villi and myometrium with placental villi attached to the <u>superficial myometrium</u> , without invasion of deep myometrium
Grade 2	Villi reaching <u>deep myometrial tissue</u> , occasionally reaching lumen of deep uterine vascular
Grade 3a	Villi reaching to or extending <u>beyond uterine serosa</u>
Grade 3b	Villi reaching the <u>bladder wall or urothelium</u>
Grade 3c	Villi invading <u>any other pelvic organ</u>

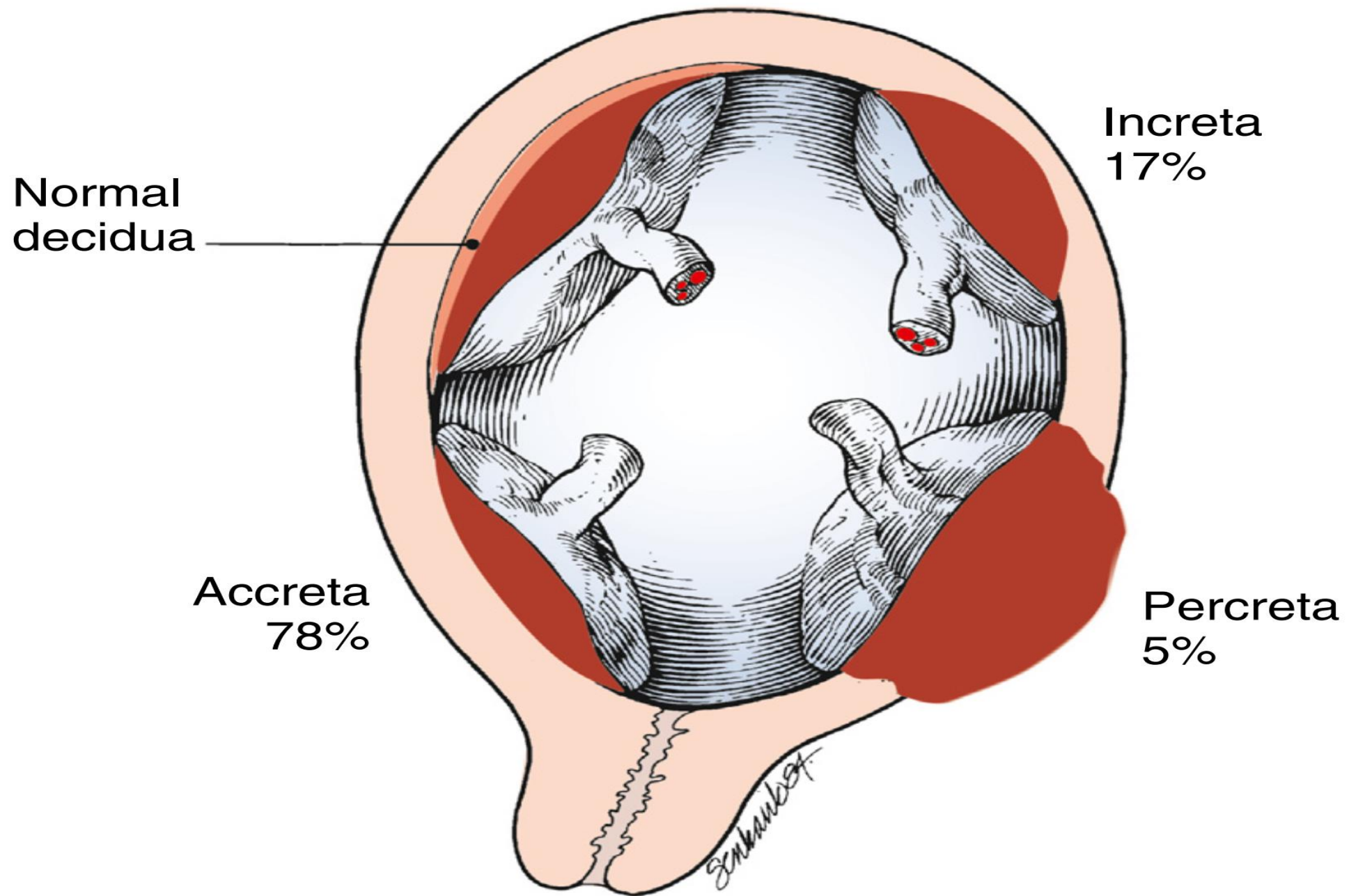


Fig. 20.7 Uteroplacental Relationships Found with Placenta Accreta Spectrum.

MANAGEMENT

1st TM PAS

2ND TM PAS

3rd TM PAS

TABLE 1**International Society for Abnormally Invasive Placenta (IS-AIP) criteria for what constitutes a specialist center for AIP**

1. A center that can provide a multidisciplinary team (MDT) with significant experience in managing abnormally invasive placenta (AIP) and that can provide antenatal diagnosis and preoperative planning. This team should be available 24 hours a day, 7 days a week, to ensure that expertise is available for emergency situations.
This MDT should, as a minimum, include:
 - Imaging expert (fetal medicine specialist and/or radiologist)
 - Experienced obstetrician (often maternal-fetal medicine specialist)
 - Anesthesiologist with expertise in complex obstetric cases
 - Surgeon experienced with complex pelvic surgery (often a gynecological oncologist)
 - Urologist (with experience of open urological surgery especially ureteric re-implantation)
 - Neonatologist
 - Interventional radiologist^a
2. There should be, *on site*, rapid access to the following in case of emergency:
 - Colorectal surgeon
 - Vascular surgeon
 - Hematologist
3. Adult intensive care facilities available on site
4. Gestational age—appropriate neonatal intensive care facilities
5. Massive transfusion facilities
6. Intraoperative blood salvage (cell salvage) services available^b

1ST TM PAS MANAGEMENT

Medical intervention	Surgical Intervention	Radiologic Intervention
• Intra gestational MTX injection • Intra gestational KCl injection • Trans cervical insertion of balloon catheters • Systemic MTX?	• US-guided/Laparoscopic suction aspiration • Operative resection • Hysterectomy	• UAE

2ND TMP PAS MANAGEMENT

- Although the impact of PAS on pregnancy outcomes is well described,
 - No RCT have examined the management of PAS in the 2nd TM.
- The treatment approach is based on a number of factors, and should be individualized.
- The patient needs to be counselled extensively about:
 - Treatment options and
 - The risk of recurrent placental abnormalities in future pregnancies.
- There are two management approaches for women with suspected PAS in the 2nd TM:
 - A. Pregnancy preservation
 - B. Pregnancy termination

A. PREGNANCY PRESERVATION

- When there's no indication for pregnancy termination.
- All patients with suspected PAS should be counselled about the dx & potential sequelae.
 - Prenatal care
 - Hospitalization
 - Antenatal corticosteroids

B. PREGNANCY TERMINATION

- Obstetric/medical indication for pregnancy termination
- Informed consent
- Multidisciplinary care team

ACOG-2018: When the diagnosis of PAS is made in the previable period:

- Counselling about the **possibility of pregnancy termination for maternal indications** given the significant risks of maternal morbidity & mortality.
 - There are currently limited data to support the magnitude of risk reduction,
 - Pregnancy termination in the setting of suspected PAS also carries risk.

Cont'd

- Depending on the degree of invasion and the patient's future reproductive plans, suspected PAS is often managed with:
 1. Gravid hysterectomy
 2. Double set-up D&E
 3. Medical induction
 4. Hysterotomy
- Women wishing for fertility preservation should be counselled that their risk of PAS in a subsequent pregnancy is **22—29%**.

3RD TM PAS MANAGEMENT

A. Conservative management:

- Eventhough safety remains uncertain conservative management seems like an attractive option because:
 - It **avoids a morbid surgery** and
 - It may lead to **preserved fertility**.
- Conservative management should be undertaken with:
 - Extreme caution and
 - Appropriate counselling regarding risks.

B. Definitive management:

- **Cesarean hysterectomy:**

Surgical risks ↑ with the depth of placental invasion. Women with placenta percreta are more likely to

 - Require additional blood products,
 - Have urologic injury, and
 - Require ICU admission than women with simple placenta accreta.
 - Blood products, including RBCs, FFP, & platelets

Placenta accreta spectrum (PAS)

Prenatal care

Counsel regarding risks
Correct nutritional anemia
Antenatal corticosteroids
Consider magnesium neuroprotection

Delivery preparation

Multidisciplinary consultation
Appropriate IV access
Blood products available
DVT prophylaxis
Bladder catheter +/- ureteral stents
Internal iliac artery balloons/catheters

Delivery

Cesarean hysterectomy

Conservative management

Postpartum

ICU care

Complications

Acute complications:

- The main risk associated with the delivery of a woman with PAS:
 - MOH
 - Massive transfusion,
 - Coagulopathy,
 - Urinary tract injury, and

Long term complications:

- Successful conservative treatment for PAS doesn't appear to compromise subsequent fertility or obstetrical outcome, but a pregnancy following a previous PAS is at **↑d risk for adverse maternal outcomes**, including:
 - Recurrent PAS: 22-29%
 - Uterine rupture: 3.3%
 - PPH: 8.6%
 - Peripartum hysterectomy: 3.3%
 - Blood transfusions is 16.7%
 - Uterine synechiae with 2^o-amenorrhea

SUMMARY

- Maternal & fetal outcomes with PAS have improved considerably with antenatal diagnosis and modern management.
- Those women with PAS should be managed at a center with a MDT & experience in treating placenta accreta.
- Given the lack of RCT, the optimal management of PAS remains undefined.
- The approach should be individualized & determined in collaboration with the patient.

REFERENCES

- Williams Obstetrics 26th edition
- Gabbe's Obstetrics 9th edition

THANK YOU